Fiscal Year:	FY 2004	**Task Last Updated:**	FY 07/08/2004
PI Name:	Townsend, Lawrence W. Ph.D.		
Project Title:	Development of a Monte Carlo Radiation Transport Code System for HEDS		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	865-974-7569
Organization Name:	University of Tennessee		
PI Address 1:	Nuclear Engineering Department		
PI Address 2:	University of Tennessee		
PI Web Page:	Aspects of the FLUKA code development work are available at http://fleur.cern.ch/		
City:	Knoxville	**State:**	TN
Zip Code:	37996-2300	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	08/01/2000	**End Date:**	11/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Barghouty, N	**Contact Phone:**	256 544-0238
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Gabriel, T. (Independent Consultant)		
Grant/Contract No.:	NAG8-1669		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Develop a radiation transport code system for HEDS applications, based on Monte Carlo methods, that is state-of-the-art in terms of physics models, transport methodologies, and capabilities. To develop a radiation transport code system for HEDS. This transport method (Monte Carlo) has inherent capabilities – namely 3D transport, modular programming and detailed output – that are important for HEDS applications. The new radiation transport code will provide a near-term computational tool with reduced uncertainties for HEDS radiation issues.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	HETC code can also be used for accelerator beam characterization and shielding studies.		

Task Progress:	1. The HETC code has been modified for heavy ion transport by devveloping and incorporating the appropriate transport modules into the code. The initial heavy ion event generator incorporating the nuclear models used to describe the interactions and breakup of the incident beam nuclei has been developed and incorporated into the HETC code. Code validation has been initiated by comparing predicted fluences of heavy ion species (incident beams and secondaries) with laboratory measurements obtained by NASA funded research studies at the Brookhaven National Laboratory. Agreement between code predictions and laboratory measurements for fragment fluences near the incident beam direction is good for targets as thick as 10 g/cm2. There are some increasing differences noted at 10 g/cm2, but further comparisons with other data are needed to establish if the disagreement is due to data errors or problems with the code predictions. Code testing and refinement are continuing under the auspices of NASA Grant No. NAG 8-1901, which continues the work begun under the current task. 2. A University of Tennessee Ph.D. student, funded for the past 2 years by a NASA Graduate Student Researcher Program Fellowship, has undertaken the development of the heavy ion event generator as part of his Ph.D. dissertation requirements, under the supervision of the PI. The initial version of the event generator has been developed, debugged, and incorporated into the HETC code. Code and event generator testing, validation and benchmarking has begun. The main shortcomings in the prersent event generator are the need for an improved treatment of secondary nucleon production and the need to incorporate meson production from nucleus-nucleus collisions. 3. Further work on this task is being carried out under the auspices of NASA Grant No. NAG 8-1901.
Bibliography Type:	Description: (Last Updated: 06/13/2007)
Articles in Peer-reviewed Journals	Miller, T. M. and Townsend, L. W. "Double Differential Light Ion Production Cross Sections" Radiation Protection Dosimetry (accepted for publication) , Mar-2004
Articles in Peer-reviewed Journals	Miller, T. M. and Townsend, L. W. "Double Differential Heavy Ion Production Cross Sections" Radiation Protection Dosimetry (accepted for publication). , Mar-2004
Articles in Peer-reviewed Journals	Miller, T. M. and Townsend, L. W. " Comprehensive Cross Section Database Development for Generalized Three Dimensional Radiation Transport Codes" Nuclear Science and Engineering (accepted for publication). , May-2004
Articles in Peer-reviewed Journals	Townsend, L. W.; Miller, T. M.; and Gabriel, T.A. "HETC Radiation Transport Code Development for Cosmic Ray Shielding Applications in Space" Radiation Protection Dosimetry (submitted May 2004). , May-2004
Presentation	Townsend, L. W. "Implications of the Space Radiation Environment for Human Exploration in Deep Space (Invited Plenary Presentation)." 10th International Conference on Radiation Shielding/13th ANS Topical Meeting on Radiation Protection and Shielding, Funchal, Portugal, May 9-14, 2004. May-2004
Presentation	Miller, T. M. and Townsend, L. W. "Double-Differential Light Ion Production Cross Sections" Ninth Symposium on Neutron Dosimetry, NEUDOS 9, Delft, The Netherlands, September 28 - October 3, 2003. Sep-2003
Presentation	Miller, T. M. and Townsend, L. W. "Double-Differential Heavy Ion Production Cross Sections" Ninth Symposium on Neutron Dosimetry, NEUDOS 9, Delft, The Netherlands, September 28 - October 3, 2003. Sep-2003
Presentation	Townsend, L. W.; Miller, T. M.; and Gabriel, T. A. "HETC Radiation Transport Code Development for Cosmic Ray Shielding Applications in Space" 10th International Conference on Radiation Shielding/13th ANS Topical Meeting on Radiation Protection and Shielding, Funchal, Portugal, May 9-14, 2004. May-2004
Presentation	Miller, T. M.; Townsend, L. W.; Gabriel, T. A.; Handler, T. "Extension of the HETC Radiation Transport Code to Include HZE Particle Transport" 3rd International Workshop on Space Radiation Research (IWSRR) and 15th Space Radiation Health Investigator’s Workshop, Port Jefferson, NY, May 16-19, 2004. May-2004